

NETWORK RECONNAISSANCE & PORT SCANNING AUDIT

Task 03 — Authorized Security Lab

RabTech Academy — Cybersecurity Internship

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Assessment Environment: LAB-02 — Deliberately Vulnerable VM

Target: 192.168.56.20

Tools: Nmap and Wireshark

Date: 11.09.2026

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1. Objective

The objective of this assessment was to perform authorized network reconnaissance and port-scanning against the deliberately vulnerable LAB-02 virtual machine. The assessment focused on identifying reachable services, service versions, operating-system information, UDP exposure, and relevant network packets.

2. Authorized Scope

Field	Authorized Value
Asset ID	LAB-02
Asset Name	Deliberately vulnerable VM
Owner	RabTech lab
Environment	Private host-only network
Target IP	192.168.56.20
Testing	Authorized
Excluded Actions	No persistence or outbound traffic

3. Rules of Engagement

- Testing was limited to the explicitly authorized LAB-02 target.
- No denial-of-service or destructive activity was performed.
- No persistence was attempted.
- No third-party systems were scanned.
- Evidence was limited to the minimum required for the assessment and report.

4. Methodology

The following authorized activities were performed:

- Target reachability validation using ICMP.
- TCP SYN reconnaissance using Nmap.
- Service and version detection using Nmap.
- Operating-system detection using Nmap.
- UDP reconnaissance against the top 100 UDP ports.
- Packet analysis in Wireshark during controlled Nmap scans.

5. Target Reachability

Command executed:

```
ping -c 4 192.168.56.20
```

```
L$ ping -c 4 192.168.56.20
PING 192.168.56.20 (192.168.56.20) 56(84) bytes of data.
64 bytes from 192.168.56.20: icmp_seq=1 ttl=64 time=1.02 ms
64 bytes from 192.168.56.20: icmp_seq=2 ttl=64 time=1.51 ms
64 bytes from 192.168.56.20: icmp_seq=3 ttl=64 time=1.10 ms
64 bytes from 192.168.56.20: icmp_seq=4 ttl=64 time=1.34 ms

— 192.168.56.20 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.016/1.239/1.505/0.193 ms
```

Figure 1: Target Reachability

Key result: Target 192.168.56.20 was reachable, with 4 packets transmitted and 4 packets received.

6. Nmap Reconnaissance Results

6.1 TCP SYN Scan

Command executed:

```
sudo nmap -sS 192.168.56.20
```

Key result: TCP SYN scan identified ports 22/tcp (SSH) and 80/tcp (HTTP) as open.

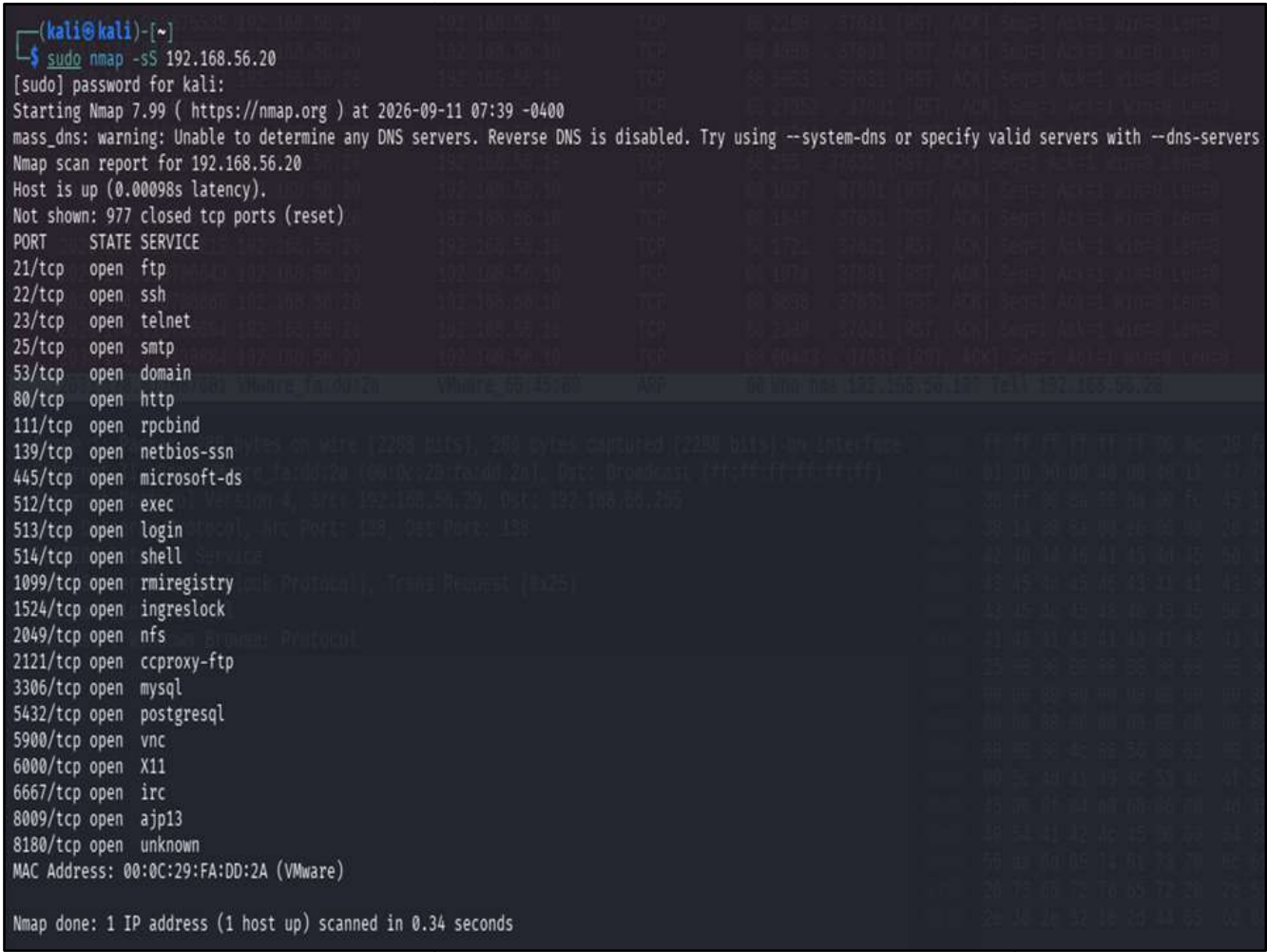


Figure 2: TCP SYN Scan

Key result: TCP SYN scan identified ports 22/tcp (SSH) and 80/tcp (HTTP) as open.

6.2 Service and Version Detection

Command executed:

```
sudo nmap -sV 192.168.56.20
```

```
(kali@kali)-[~]
└─$ sudo nmap -sV 192.168.56.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-11 07:42 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.20
Host is up (0.0020s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rexecd
513/tcp   open  login          OpenBSD or Solaris rlogind
514/tcp   open  shell          Netkit rshd
1099/tcp  open  java-rmi       GNU Classpath grmiregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  X11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:FA:DD:2A (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.01 seconds
```

Figure 3: Service Version Detection

Key result: Nmap identified numerous open services and their versions on the target. Notable services included **FTP**, **SSH**, **Telnet**, **SMTP**, **HTTP**, **SMB**, **MySQL**, **PostgreSQL**, **VNC**, and **Tomcat**, indicating a broad exposed service surface for further security assessment.

6.3 Operating-System Detection

Command executed:

```
sudo nmap -O 192.168.56.20
```

```
(kali@kali)-[~]
└─$ sudo nmap -O 192.168.56.20
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-11 07:48 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.20
Host is up (0.0015s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc (RFB Protocol), Trans Request (RFB)
6000/tcp  open  X11 (RFB)
6667/tcp  open  irc (RFB Protocol)
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:FA:DD:2A (VMware)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.63 seconds
```

Figure 4: OS Detection

Key result: Nmap OS detection identified the target system as Linux 2.6.x, with the OS details indicating Linux kernel versions 2.6.9–2.6.33. The target was one network hop away from the scanning machine.

6.4 UDP Scan

Command executed:

```
sudo nmap -sU --top-ports 100 192.168.56.20
```

```
(kali@kali)-[~] protocol, Src Port: 130, Dst Port: 130
$ sudo nmap -sU --top-ports 100 192.168.56.20
[sudo] password for kali: (Blank Protocol), Trans Request (8x25)
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-11 08:20 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-
Nmap scan report for 192.168.56.20
Host is up (0.0012s latency).
Not shown: 50 closed udp ports (port-unreach), 46 open/filtered udp ports (no-response)
PORT      STATE SERVICE
53/udp    open  domain
111/udp   open  rpcbind
137/udp   open  netbios-ns
2049/udp  open  nfs
MAC Address: 00:0C:29:FA:DD:2A (VMware)

Nmap done: 1 IP address (1 host up) scanned in 48.52 seconds
```

Figure : UDP Scan

Key result: The UDP scan identified **4 open UDP services** on LAB-02: **53/UDP (DNS)**, **111/UDP (rpcbind)**, **137/UDP (NetBIOS-NS)**, and **2049/UDP (NFS)**. The scan also showed **50 closed** and **46 open|filtered** UDP ports, indicating additional network services that may require further validation.

7. Open Ports and Services

Record only the ports/services actually reported by Nmap.

Port	Protocol	State	Service	Version
21	TCP	Open	FTP	vsftpd 2.3.4
22	TCP	Open	SSH	OpenSSH 4.7p1 Debian 8ubuntu1
23	TCP	Open	Telnet	Linux telnetd
25	TCP	Open	SMTP	Postfix smtpd
53	TCP	Open	Domain	ISC BIND 9.4.2

80	TCP	Open	HTTP	Apache httpd 2.2.8 (Ubuntu) DAV/2
111	TCP	Open	rpcbind	2 (RPC #100000)
139	TCP	Open	netbios-ssn	Samba smbd 3.X-4.X
445	TCP	Open	netbios-ssn	Samba smbd 3.X-4.X
512	TCP	Open	exec	netkit-rsh rexecd
513	TCP	Open	login	OpenBSD or Solaris rlogind
514	TCP	Open	shell	Netkit rshd
1099	TCP	Open	java-rmi	GNU Classpath grmiregistry
1524	TCP	Open	bindshell	Metasploitable root shell
2049	TCP	Open	nfs	2-4 (RPC #100003)
2121	TCP	Open	ftp	ProFTPD 1.3.1
3306	TCP	Open	mysql	MySQL 5.0.51a-3ubuntu5
5432	TCP	Open	postgresql	PostgreSQL 8.3.0-8.3.7
5900	TCP	Open	vnc	VNC (protocol 3.3)
6000	TCP	Open	X11	access denied
6667	TCP	Open	irc	UnrealIRCd
8009	TCP	Open	ajp13	Apache Jserv (Protocol v1.3)
8180	TCP	Open	http	Apache Tomcat/Coyote JSP engine 1.1

8. Consolidated Nmap Evidence

Command executed:

sudo nmap -sS -sV -O 192.168.56.20 -oN task3-nmap.txt

```

(kali@kali)-[~]
└─$ sudo nmap -sS -sV -O 192.168.56.20 -oN task3-nmap.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-11 08:25 -0400
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.20
Host is up (0.0012s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet        Linux telnetd
25/tcp    open  smtp          Postfix smtpd
53/tcp    open  domain        ISC BIND 9.4.2
80/tcp    open  http          Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind       2 (RPC #100000)
139/tcp   open  netbios-ssn   Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn   Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec          netkit-rsh rexecd
513/tcp   open  login         OpenBSD or Solaris rlogind
514/tcp   open  shell         Netkit rshd
1099/tcp  open  java-rmi      GNU Classpath grmiregistry
1524/tcp  open  bindshell     Metasploitable root shell
2049/tcp  open  nfs           2-4 (RPC #100003)
2121/tcp  open  ftp           ProFTPD 1.3.1
3306/tcp  open  mysql         MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql    PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc           VNC (protocol 3.3)
6000/tcp  open  X11           (access denied)
6667/tcp  open  irc           UnrealIRCd
8009/tcp  open  ajp13         Apache Jserv (Protocol v1.3)
8180/tcp  open  http          Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:FA:DD:2A (VMware)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.76 seconds

(kali@kali)-[~]
└─$ cat task3-nmap.txt
# Nmap 7.99 scan initiated Fri Sep 11 08:25:46 2026 as: /usr/lib/nmap/nmap -sS -sV -O -oN task3-nmap.txt 192.168.56.20
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.20

```

Figure 6: Consolidated Nmap Scan

9. Wireshark Packet Analysis

9.1 TCP SYN Traffic

Wireshark was used to observe TCP traffic generated during the controlled Nmap scan. The following display filter was used:

```
ip.addr == 192.168.56.20 && tcp.flags.syn == 1
```

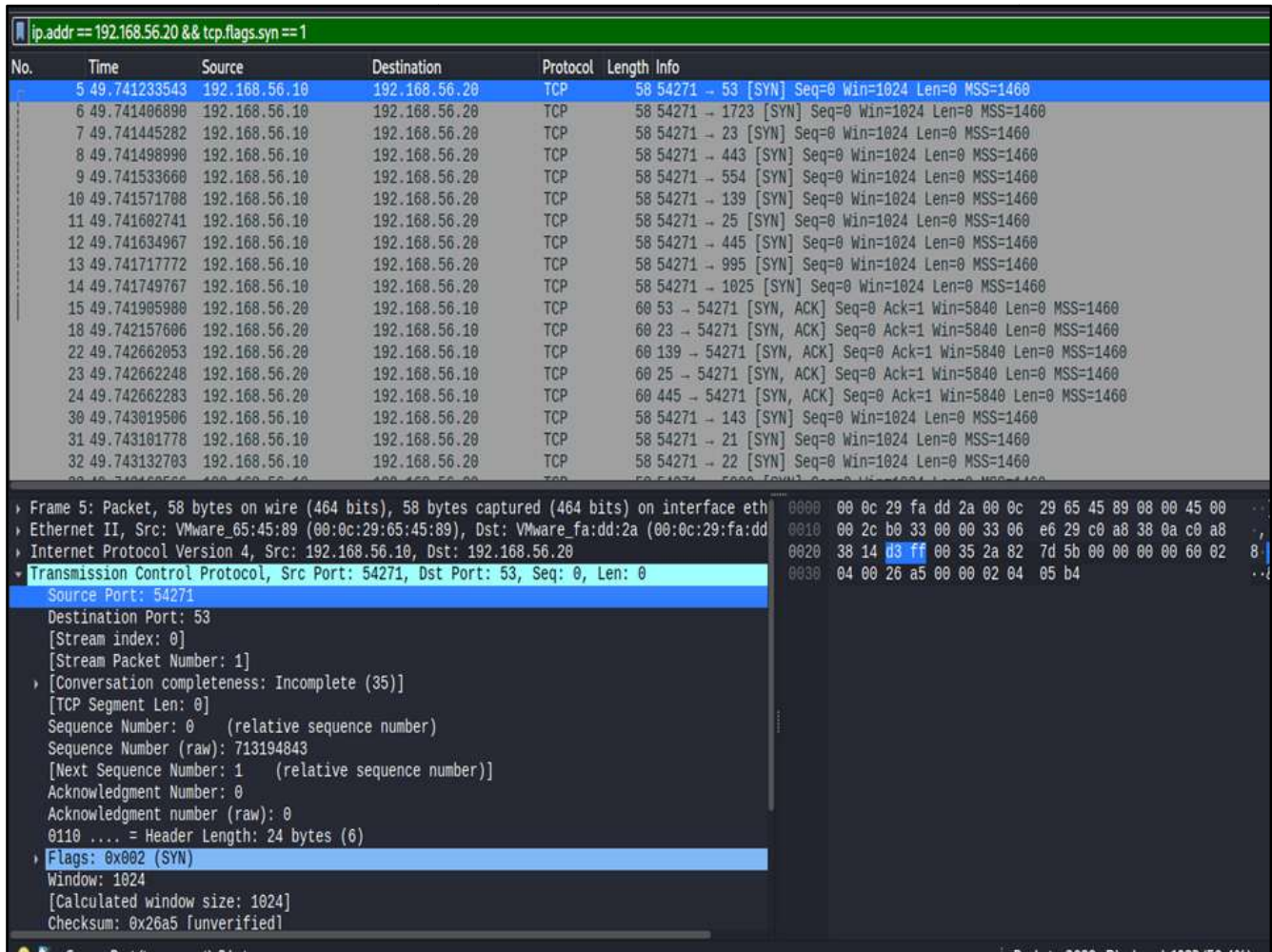


Figure7: Wireshark Capture of TCP SYN Reconnaissance Traffic

Observation: Wireshark captured TCP SYN packets from 192.168.56.10 to 192.168.56.20 across multiple destination ports. The packets show the SYN flag, indicating connection initiation attempts during the TCP reconnaissance scan. The traffic confirms that the scanner was probing multiple TCP services.

9.2 UDP Traffic

A controlled UDP scan was observed using Wireshark.

```
ip.addr == 192.168.56.20 && udp
```

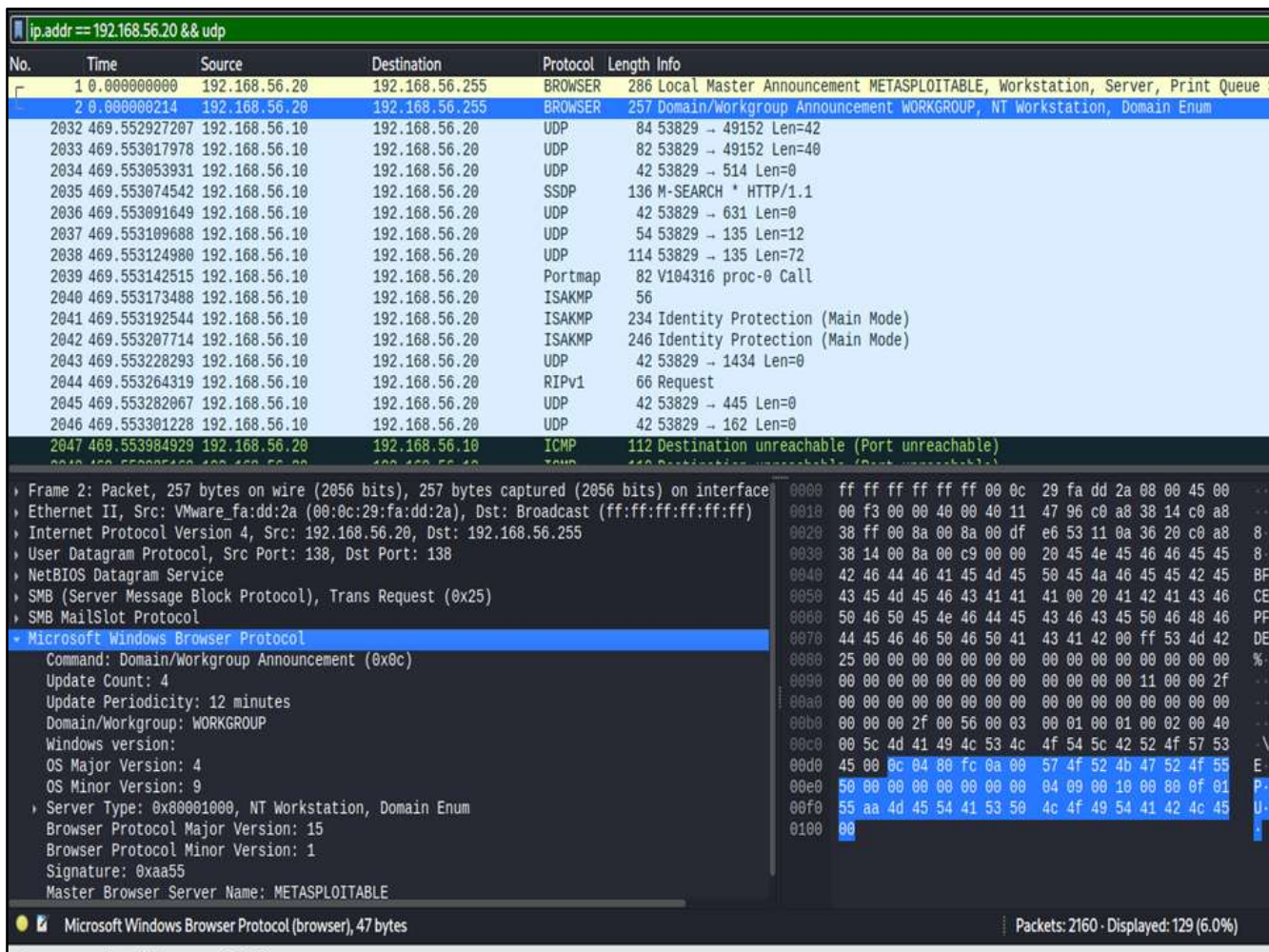


Figure 8: Wireshark Capture of UDP Reconnaissance Traffic and ICMP Port-Unreachable Response

Observation: Wireshark captured UDP traffic between 192.168.56.10 and the authorized target 192.168.56.20. The capture shows UDP probes to multiple destination ports, including ports associated with SMB/NetBIOS and other services. An ICMP “Destination unreachable (Port unreachable)” response is also visible, indicating that at least some probed UDP ports were closed.

10. Conclusion

The Network Reconnaissance and Port Scanning Audit successfully assessed the authorized LAB-02 deliberately vulnerable virtual machine (192.168.56.20) using Nmap and Wireshark. The reconnaissance identified a large number of exposed TCP and UDP services, including FTP, SSH, Telnet, SMTP, HTTP, SMB/NetBIOS, DNS, NFS, MySQL, PostgreSQL, VNC, IRC, and Tomcat. Service and version enumeration also revealed several legacy software versions, demonstrating a broad attack surface within the intentionally vulnerable training environment. The UDP assessment identified open services such as DNS, rpcbind, NetBIOS-NS, and NFS, while Wireshark packet analysis provided network-level evidence of the reconnaissance activity. TCP SYN packets demonstrated connection-initiation probes against multiple ports, and UDP traffic showed probes followed by ICMP port-unreachable responses for closed services. Overall, the assessment demonstrates why network reconnaissance, service enumeration, and packet-level analysis are important stages of a security assessment. The results should be treated as reconnaissance findings rather than proof that every identified service is exploitable. In a real production environment, unnecessary services should be disabled, exposed services should be restricted through firewall rules, legacy software should be upgraded, and network traffic should be monitored for scanning activity. All testing in this report was performed within the authorized isolated training environment and within the defined rules of engagement.

Prepared by Pritam Bose

11. References

1. Nmap — Network Mapper
Official documentation and reference for TCP SYN scanning, UDP scanning, service/version detection, and OS detection.
[Nmap Official Documentation](#)
2. Wireshark Documentation
Official documentation for packet capture, filtering, and network traffic analysis.
[Wireshark User's Guide](#)
3. Wireshark Display Filters
Reference for packet filtering syntax used during traffic analysis.
[Wireshark Display Filter Reference](#)
4. Metasploitable 2 Documentation / Training Reference
Background reference for the intentionally vulnerable Linux training environment used as LAB-02.
5. RabTech Lab — Authorized Security Lab Rules of Engagement
Provided authorization, asset scope, permitted activities, exclusions, evidence-handling requirements, and reporting requirements for this assessment.

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